

Case Study: Loan Portfolio Quality Analysis

Candidate Instructions & Technical Guidelines

1. Objective

Analyze the provided lending datasets to identify risk trends, segment the portfolio, and provide actionable recommendations to the Chief Risk Officer (CRO).

2. Technical Setup

To ensure a level playing field, please use **BigQuery (Standard SQL)** or **Python (Pandas)** for your analysis.

- **If using BigQuery:** We recommend using the [BigQuery Sandbox](#), which is free and does not require a credit card.
- **Data Loading:** Create a dataset and upload [loans_data.csv](#) and [payments_data.csv](#).
- **Note on Data Quality:** Real-world data is often "noisy." If you encounter issues with timestamp precision or malformed strings during ingestion or querying, we expect you to handle these programmatically within your scripts.

3. Key Business Rules

- **Disbursement Cost:** Every loan has a fixed cost of **KES 70** at the time of disbursement.
- **Default Definition:** A default occurs the moment a customer misses a full payment on their `loan_due_date`.
- **Write-Off Policy:** Loans are written off 120 days after the `loan_due_date` if they remain unpaid.
- **Cohorts:** Use the **Disbursement Month** as the primary cohort for tracking performance over time.

4. Deliverables

Please submit a folder containing the following:

1. **The Code:** Your SQL scripts (`.sql`) or Python Notebook (`.ipynb`). Your code should be clean, commented, and runnable.
2. **Visualizations:** Any charts or dashboards created (PNG, PDF, or a link to a BI tool like Looker Studio/Tableau).

5. Evaluation Criteria

- **Data Accuracy:** Correct calculation of PAR (Portfolio at Risk), Recovery Rates, and NIM.
- **Analytical Depth:** Ability to identify high-risk segments beyond simple averages (e.g., using Cohort/Vintage analysis).
- **Technical Proficiency:** How you handle data types, joins, and edge cases (like the nanosecond precision issue).
- **Communication:** Translating complex data into clear, executive-level recommendations.

Case Study: Loan Portfolio Quality Analysis for a Lending Business

Objective: Analyze the quality of a lending portfolio to identify key risks, trends, and opportunities for improved lending decisions.

Part 1: The Scenario & Your Role (Introduction)

Scenario: You are a Business Intelligence Analyst at a growing financial institution that provides various types of loans. The Chief Risk Officer (CRO) has expressed concern about the overall health of the loan portfolio. They suspect there might be pockets of higher risk or deteriorating quality, and they need a data-driven approach to understand the portfolio's performance, identify problematic trends, and potentially inform adjustments to credit policies or collection strategies.

Your Goal: The CRO has tasked you with conducting a comprehensive analysis of the current loan portfolio quality. You need to define key metrics, identify risk factors, segment the portfolio, and provide actionable insights.

1. `loans_data.csv` extract:

- `loan_code` (Unique identifier for each loan)
- `customer_id` (Unique identifier for each customer)
- `product_type` (e.g., 'Verified Sales', 'Business Loan', 'Personal Loan', 'Education Loan')
- `Disbursed_Date` (Date loan was disbursed)
- `Disbursed_amount` (Original principal amount)
- `loan_due_date` (The final expected repayment date)
- `settlementdate` (Actual date loan was fully paid, NULL if ongoing/not settled)
- `Gross_loan_interest` (Total interest charged over the loan term)
- `unearned_interest` (Interest yet to be earned on active loans)
- `surplus_payment` (Positive value if customer overpaid, negative if underpaid)
- `Penalties` (Total penalties incurred by the customer on this loan)
- `Waivers` (Total waivers applied to the loan, e.g., waiving penalties)
- `Initiation_fee` (Fees paid by new customers)
- `Admin_fee` (Fees charged at origination)
- `Appointment_fee` (Fees charged at origination)
- `Joining_fee` (Fees charged at origination)
- `fix` (Fees charged to correct any undercharge or overcharge)
- `total_due` (Total amount expected to be paid back including principal, interest, fees, penalties)

2. `payments_data.csv` extract:

- `date` (Date of individual payment)
- `loan_rep_code` (Links to `loans_data.loan_code`)

- **amount** (Amount of the individual payment)
 - **Potentially Derivable:**
 - **is_late** (Boolean: TRUE if payment was late, FALSE otherwise - requires linking to expected payment schedule)
 - **payment_type** (e.g., 'Regular', 'Early Repayment', 'Late Payment Fee')
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Part 2: Interview Questions & Tasks

Task 1: Defining Portfolio Quality Metrics

- **Question:** How would you define "loan portfolio quality"? What are the key metrics you would propose to the CRO to measure and monitor the health of the lending portfolio? Write SQL queries (or **pandas** operations) to calculate:
 - *(Expect discussion on: Default Rate, Write-off Rate (Due Date +120 days), Portfolio at Risk (PAR), Recovery Rate, Average Days Past Due (DPD), Net Interest Margin (NIM) per loan, Loan Loss Provisioning & how this affects cost)*
- **Follow-up:** How would you identify emerging trends in loan quality over time (e.g., deteriorating performance for a specific **product_type** disbursed in a particular period)? What role would **Disbursed_Date**, **settlementdate**, or **cohorts** play in this trend analysis?

Part 3: Interview Questions & Tasks (Adding Experimentation)

Task 7: Designing Experiments for Portfolio Quality Improvement

- **Question (a):** Let's say, based on your analysis, you've identified a specific segment of loans (e.g., 'Verified Sales' loans disbursed with high **initiation_fee** that tend to accrue **Penalties**) where you believe a change in policy or a targeted intervention could improve portfolio quality.
 - How would you design an A/B test (or another experimental design) to validate this hypothesis?
 - What are the key elements of your experimental design
 - **What specific metrics** from your **loans_data** and **payments_data** would you use as the **primary outcome variable(s)** to measure the success of your intervention?
 - **How would you determine statistical significance** of the results, and what statistical tests or concepts would you apply?

